VVU EARTH TECH

User Manual — Outreach Distribution Edition

FOR OUTREACH DISTRIBUTION, SCOPING & SALES FRAMEWORK STRATEGIES

Target Audience: End Users, Operators, Municipal Stakeholders

CLASSIFICATION: Outreach Distribution — Scoping & Sales Framework Strategy

Content Scope: Technical, Operational, and Strategic — Excludes Legal & Financial

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Venture Vision Ubuntu (VVU) — EARTH TECH Division

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1. What is VVU EARTH TECH?

The **VVU EARTH TECH** platform is a deterministic evidence runtime for infrastructure monitoring, municipal water network management, and autonomous decision-making. It combines an **Epistemic DAG Runtime** (with four primitives: Fact, Proof, Policy, Projection), a **Trust Runtime** (5-state AIR safety pipeline), and a **72-Hour Resilience Matrix** to deliver verifiable, replay-deterministic, append-only evidence for every decision the system makes.

Core Design Principles

- **Deterministic Replay:** Every decision in the system can be replayed from the evidence log, producing the identical result every time.
- **Append-Only Evidence:** No Fact can be modified or deleted after acceptance — the log is cryptographically sealed.
- **Golden Rule:** The AIR Kernel is horizontal infrastructure — no product-specific logic in the open-source code.
- **Fail-Closed Philosophy:** When the system cannot verify, it rejects — never accepts unverified inputs.

Four Epistemic Primitives

Primitive	Purpose	Key Property
Fact	Observed evidence from sensors, logs, or external sources	Append-only, SHA-256 hashed, MMR-indexed
Proof	Cryptographic attestation linking evidence to a decision	Ed25519/RSA-PSS-SHA256/ECDSA P-384 signed
Policy	Governance rule governing how evidence is evaluated	Versioned, bi-temporal, replay-deterministic
Projection	Derived state computed from Facts and Policies	Recomputable from evidence log alone

2. The Epistemic Runtime Dashboard

The VVU MASTER Dashboard is the primary user interface for the Epistemic DAG Runtime. It provides a unified view of all system components: trust runtime status, validation progress, resource acquisition tracking, resilience metrics, and policy governance.

Dashboard Sections

Section	Description	User Action
Overview	System health, validation index, phase progress	View real-time status
Trust Runtime	5-state AIR safety pipeline, risk scores, gates	Monitor safety state
Circuit Breaker	3-state municipal breaker + 5-state AIR breaker	Track state transitions
Resilience Matrix	72-Hour resilience pillars status	Review pillar health
Policy Studio	Policy creation, versioning, diff comparison	Create/edit policies
DAG Topology	Interactive graph of epistemic dependencies	Explore evidence graph
MMR Proofs	Merkle Mountain Range proofs and verification	Verify evidence chains
Timeline	Historical events and milestone tracking	Review event history
Validation Suite	VVU-VAL-001 phases, milestones, outreach	Track validation progress
Resource Acquisition	7-Track strategy status overview	Review outreach progress
CLI Terminal	Direct command interface (air health, air ledger)	Execute commands
Performance Metrics	System throughput, latency, queue depth	Monitor performance
Audit Reports	Evidence bundle verification and audit trail	Review audit results

3. Navigating the Dashboard

The dashboard uses a tabbed interface with keyboard shortcuts for rapid navigation. Each section loads dynamically for performance optimization. The command palette (Ctrl+K) provides instant search across all sections and commands.

Keyboard Shortcuts

Shortcut	Action
Ctrl+K	Open command palette (global search)
1-9	Switch to section tab by number
Ctrl+/	Show keyboard shortcuts overlay
Escape	Close overlay/modal

Interactive Features

- **Drag & Drop:** DAG topology nodes can be rearranged by dragging.
- **Zoom & Pan:** Graph sections support zoom (scroll) and pan (drag).
- **Click-to-Expand:** Click any node to see full detail in an overlay panel.
- **Real-Time Updates:** Validation suite and trust runtime update live during a validation event.
- **Theme Toggle:** Switch between dark and light themes (preserves readability).

4. Trust Runtime Safety Pipeline

The Trust Runtime implements a 5-state AIR (Autonomous Intelligence Runtime) safety pipeline with hysteresis to prevent oscillation at threshold boundaries. This is **separate from** the 3-state municipal infrastructure circuit breaker — they operate independently.

AIR Safety States

State	Meaning	Trigger	User Impact
NORMAL	All systems operating within safe parameters	Risk score below warning threshold	Full functionality, all features available
WARNING	Elevated risk detected — monitoring intensified	Risk score ≥ 0.6 or delta ≥ 0.05	Features available, monitoring alerts active
TRIPPED	Risk exceeds safe limits — execution halted	Risk score ≥ 0.75 or delta ≥ 0.1	Pending intents halted, new submissions queued
RECOVERY	Risk declining — minimum hold period enforced	Risk drops below trip threshold	Queued intents processed after hold period
ESCALATED	Repeated trips — system-wide escalation	3 trips within 1-hour window	All execution suspended, manual review required

Gate Pipeline Flow

- **Gate 0 — Temporal Validity:** Intent must not exceed 72-hour maximum age.
- **Gate A — Convergence:** Contracting behavior checked; divergence penalty applied.
- **Gate B — Accumulation:** Tier-weighted exposure must stay below configured ceiling.
- **Gate C — Velocity:** First-derivative (rate of change) must stay below max velocity.
- **Gate D — Acceleration:** Second-derivative (rate of rate) must stay below max acceleration.
- **Gate E — State Drift:** Current state must not drift beyond max distance from intent snapshot.
- **Composite Risk Score:** Weighted sum of all gate outputs → circuit breaker evaluation.

5. Circuit Breaker States

The system operates two independent circuit breaker systems, each serving a distinct purpose:

Breaker	States	Scope	Purpose
AIR Safety Pipeline	NORMAL / WARNING / TRIPPED / RECOVERY / ESCALATED	Trust Runtime	Autonomous decision safety — prevents unsafe AI actions
Municipal Infrastructure	NORMAL / DEGRADED / FAIL-CLOSED	Resilience Manager	Infrastructure resilience — protects water network monitoring

■ ***These two breakers are architecturally independent. The AIR breaker governs autonomous decision safety; the municipal breaker governs infrastructure resilience. They do not cascade.***

6. Validation Suite (VVU-VAL-001)

The VVU-VAL-001 is a 72-hour continuous validation protocol that subjects the Epistemic Runtime to controlled failure injection across 6 phases. The protocol is pre-registered (test plan frozen before execution) with published success criteria and a computed Validation Index (PASS ≥ 90.0).

Validation Phases

Phase	Name	Hours	Gate	Objective	Severity
P1	Nominal Load	0–12h	Baseline	Establish baseline under normal telemetry traffic	Critical
P2	Telemetry Flood	12–24 h	Acceptance Capacity	Verify acceptance pipeline absorbs 100× flood	Major
P3	Network Chaos	24–36 h	HLC Ordering	Verify replay stays deterministic under packet loss	Critical
P4	Storage Pressure	36–48 h	Append-Only Integrity	Verify graceful degradation under disk fill	Critical
P5	Node Failure	48–60 h	Recovery	Verify pods restart and no Fact is lost	Major
P6	Security Injection	60–66 h	HF-001/002/005	Verify every spoofed/malformed payload rejected	Critical
P7	Partition + Recovery	66–72 h	LVL-17	Verify deterministic HLC merge after partition	Critical

Validation Index Dimensions

Dimension	Weight	What It Measures
Replay Determinism	0.25	Checksum match between live and replayed Fact Log
Append-Only Integrity	0.20	No Fact modified or deleted; MMR root valid
Policy Enforcement	0.15	Every violation produces Failure Fact; no crash
HLC Merge Correctness	0.15	Zero conflicts on partition reconnect
Evidence Bundle Integrity	0.15	Hourly bundles SHA-256 verified
Security Gate Rejection	0.10	All spoofed/malformed payloads rejected

■ ***PASS threshold: Validation Index ≥ 90.0. Zero Critical failures required (§3.1).***

7. Resource Acquisition Strategy

7-Track Resource Acquisition & Partnership Strategy: Track A — Municipal Partnerships: Active outreach to Cape Town and progressive municipalities for pilot deployment and validation partnerships. *Status: Active Outreach* **Track B — University Research Partnerships:** Engagement with South African and international universities for collaborative research, validation observation, and academic publication. *Status: Strategy* **Track C — Industry Integration:** Partnership development with water infrastructure, IoT, and municipal technology companies for integration, supply chain, and co-development. *Status: Strategy* **Track D — Public Funding:** Application to national and provincial technology innovation funds, water sector grants, and digital infrastructure programmes. *Status: Strategy* **Track E — Private Investment:** Structured engagement with impact investors, technology venture capital, and ESG-aligned funds. *Status: Strategy* **Track F — Sponsorship & Equipment:** Outreach to hardware manufacturers, cloud providers, and technology sponsors for equipment, infrastructure, and in-kind support. *Status: Strategy* **Track G — Community & Open Source:** Building developer community, open-source contributors, and civic technology networks for grassroots validation and adoption. *Status: Strategy*

■ IMPORTANT: Tracks B–G are currently at Strategy stage. Only Track A (Cape Town outreach) is at Active Outreach stage. No track has reached Confirmed Commitment. VVU presents these as planned and active initiatives, NOT as outcomes already secured.

8. Cape Town Pilot Municipality

Cape Town serves as the alternative pilot municipality case study for the VVU EARTH TECH Hydro-Gateway deployment. Cape Town was selected following a governance review that identified operational risks at the originally proposed municipality. Cape Town's established water management infrastructure, progressive governance framework, and existing IoT deployment experience make it an ideal partner for validating municipal-grade infrastructure monitoring technology.

Cape Town Selection Criteria

- **Established Water Management:** Cape Town has extensive water infrastructure experience following the 2018 water crisis, including advanced demand management and monitoring systems.
- **Progressive Governance:** Transparent governance framework with established IoT deployment experience and digital transformation initiatives.
- **Technical Infrastructure:** Existing sensor networks, data platforms, and municipal IT systems suitable for Hydro-Gateway integration.
- **Research Ecosystem:** Proximity to University of Cape Town, Stellenbosch University, and CSIR for collaborative validation.

9. Governance Framework

Execution Principle: VVU EARTH TECH advances through multiple independent pathways in parallel. Every activity is subject to formal governance review. Progress is measured by verified engineering deliverables, not announcements. Where a pathway encounters obstacles, alternative pathways continue without delay. This parallel execution model ensures continuous forward progress regardless of individual pathway outcomes.

Communications Policy: VVU EARTH TECH follows a staged disclosure framework: (1) Engineering progress is communicated only after verification — not before. (2) Outreach is structured in three stages: Evidence Publication → Personalized Outreach → General Social & Press. (3) Sensitive operational details are shared on a need-to-know basis. (4) Mass-blast communication is structurally impossible — the outreach engine enforces sequential stage gates.

Staged Release Enforcement

Stage	Name	Trigger	Audiences	Cooldown
1	Evidence Publication	M72 milestone + Validation Index ≥ 90.0	Technical communities, researchers	0 hours
2	Personalized Outreach	Stage 1 complete + 24h elapsed + no SEV-1	Investors, municipalities	24 hours
3	General Social & Press	Stage 2 complete + 48h elapsed	Journalists, general public	48 hours

10. Getting Started

Accessing the Dashboard

The VVU MASTER Dashboard is accessible via the web interface. All sections load dynamically for optimal performance. The dashboard supports both SIM (simulation) and LIVE modes for validation event monitoring.

First Steps

- **1. Open the Dashboard:** Navigate to the VVU MASTER Dashboard URL.
- **2. Review Overview:** Check system health, current phase, and validation index.
- **3. Explore Sections:** Use tab navigation or Ctrl+K to jump between sections.
- **4. Monitor Trust Runtime:** Watch the AIR safety state and risk score trends.
- **5. Track Validation:** During a VVU-VAL event, monitor phase progress and milestones.

Support & Outreach

For partnership inquiries, pilot deployment discussions, or technical questions, contact the VVU EARTH TECH outreach team. All outreach follows the staged release enforcement protocol — no unsolicited mass communication.